FICO Score Bucketing for Mortgage Default Risk

Quant Internship – Credit Risk Modeling (Task 4)



Business Context: Mortgage Portfolio Risk



Portfolio Risk Assessment

Evaluate and manage the inherent risk within the mortgage portfolio.



Estimate Probability of Default (PD)

Develop robust models to quantify the likelihood of loan default for individual mortgages.



Categorical Model Inputs

Ensure model architecture accommodates categorical variables for enhanced stability and interpretability.

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Problem Definition: FICO Score to Credit Rating



Objective:

Map a continuous FICO score and default indicator to a discrete credit rating bucket.

Key Constraints:

- Contiguous and exhaustive FICO ranges.
- Lower rating implies superior credit quality.
- Monotonic decrease in Probability of Default (PD) across ratings.

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Strategic Advantages of FICO Score Bucketing

Continuity to Discreteness

Transforms continuous FICO scores (300–850) into manageable discrete categories.

Enhanced Interpretability

Simplifies risk communication and understanding for stakeholders.

Improved Model Stability

Reduces sensitivity to minor FICO fluctuations, ensuring consistent model performance.

Regulatory Acceptance

Aligns with regulatory requirements for transparent and explainable credit risk models.

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Likelihood-Based Bucketing: Core Concept

Bernoulli Process Assumption

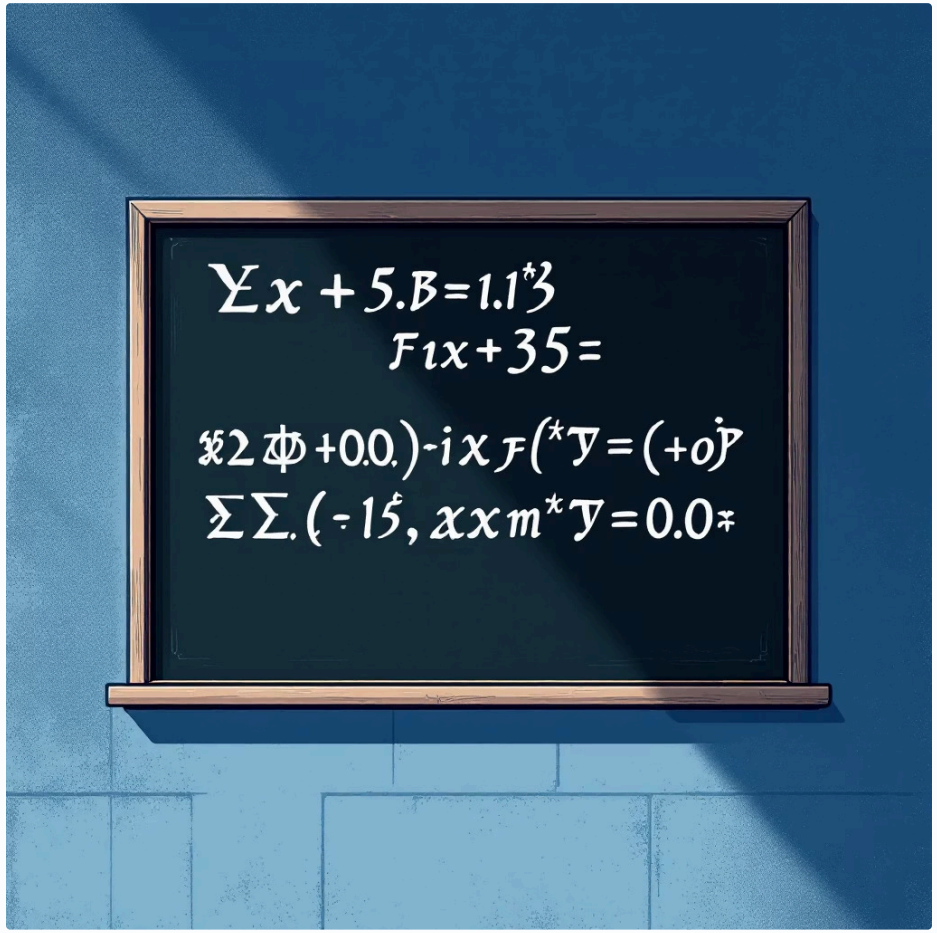
Each FICO bucket is modeled as a Bernoulli trial, representing the likelihood of default.

PD Estimation

Probability of Default (PD) for each bucket is derived from the ratio of observed defaults to total observations within that bucket.

Objective Function

Maximize the aggregate log-likelihood across all buckets to identify optimal FICO breakpoints.



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Optimization Method: Dynamic Programming

1

FICO Score Aggregation

Efficiently groups FICO scores to streamline the optimization process.

2

Dynamic Programming

Applies dynamic programming algorithms to precisely determine optimal FICO score boundaries.

3

One-Dimensional Segmentation

Focuses on segmenting FICO scores along a single dimension to create distinct risk categories.

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Derived Rating Map and PD Allocation

1	408–520	66%
2	521–580	38%
3	581–640	20%
4	641–696	10%
5	697–850	4.6%

The table illustrates the finalized FICO score rating map, showing distinct PDs for each segment.

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Interpretation: Clear Risk Separation



Distinct Credit Risk Profiles

Each bucket clearly delineates differing levels of credit risk, enhancing analytical precision.



Inverse Relationship

Demonstrates a consistent inverse relationship: higher FICO scores correlate with lower Probability of Default.



Stability & Interpretability

The derived buckets offer both predictive stability and straightforward interpretability for risk management.

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Conclusion: Towards an Integrated PD Pipeline

1

Categorical PD Modeling

Bucketing directly facilitates the integration of FICO scores into categorical PD models.

2

Generalizable Approach

The methodology is adaptable and scalable for diverse future mortgage portfolios.

3

Integration Ready

The developed framework is prepared for seamless integration into the existing mortgage PD pipeline.

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